

Project Plan: WBS, Schedule, and Risks (Entregable 1)

Team 2

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Tuesday October 13, 2026

Integridad de Seguridad Informática de Redes y Sistemas de Software

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Work breakdown structure (WBS) of the chosen project

The project is broken down into nine phases. Each activity is the smallest unit of work that is estimated, scheduled, and assigned to one team member.

The Work column shows the effort for each activity in hours (1 day = 8 working hours). For each phase, it shows the total work, which is the sum of the hours of all its activities. This total is different from the phase duration in Microsoft Project, because the duration shows how many calendar days the phase lasts, and several team members work on the same phase at the same time. We use the work because it shows how much effort each phase really needs.

The complete WBS, with the hierarchical numbering of every activity, is shown in the Microsoft Project task list below:

WBS	Task Name	Work
1	**ISSSTE**	**419.92 hrs**
1.1	**Technical**	**81.2 hrs**
1.1.1	Configure the repository and branching strategy	2.96 hrs
1.1.2	Configure the frontend	5.04 hrs
1.1.3	Configure the backend and API	5.04 hrs
1.1.4	Design the entity-relationship model	8 hrs
1.1.5	Implement the MySQL database	8 hrs
1.1.6	Implement authentication and sessions	8 hrs
1.1.7	Implement role-based authorization	13.04 hrs
1.1.8	Design the user interface and prototypes	8 hrs
1.1.9	Implement responsive and accessible navigation	8 hrs

1.1.10	Implement secure error handling	5.04 hrs
1.1.11	Configure backup and recovery mechanisms	5.04 hrs
1.1.12	Prepare synthetic test data	5.04 hrs
1.2	**Child Registry Management**	**31.12 hrs**
1.2.1	Implement child registration	8 hrs
1.2.2	Implement child search	5.04 hrs
1.2.3	Implement child information updates	5.04 hrs
1.2.4	Implement transfers between EBDIs	8 hrs
1.2.5	Implement monthly registry updates	5.04 hrs
1.3	**Parent Information and Target Height**	**10.96 hrs**
1.3.1	Register parents' height	2.96 hrs
1.3.2	Calculate family target height	8 hrs
1.4	**Anthropometric Measurements**	**29.04 hrs**
1.4.1	Implement measurement registration	8 hrs
1.4.2	Calculate the child's age	2.96 hrs
1.4.3	Validate anthropometric measurements	5.04 hrs
1.4.4	Control 45-day measurement intervals	5.04 hrs
1.4.5	Allow additional measurements	2.96 hrs
1.4.6	Display measurement history	5.04 hrs

1.5	**Growth Assessment**	**48 hrs**
1.5.1	Calculate BMI	2.96 hrs
1.5.2	Assess weight-for-height	8 hrs
1.5.3	Assess height-for-age	8 hrs
1.5.4	Assess BMI-for-age	8 hrs
1.5.5	Assess head circumference	8 hrs
1.5.6	Classify assessment results	8 hrs
1.5.7	Record the clinical reference used	5.04 hrs
1.6	**Alerts and Follow-Up**	**36.16 hrs**
1.6.1	Generate clinical alerts	8 hrs
1.6.2	Generate overdue measurement alerts	5.04 hrs
1.6.3	Register health conditions	5.04 hrs
1.6.4	Mark cases for follow-up	2.96 hrs
1.6.5	Manage follow-up statuses	5.04 hrs
1.6.6	Register follow-up actions	5.04 hrs
1.6.7	Display follow-up cases	5.04 hrs
1.7	**Regionalization**	**44.16 hrs**
1.7.1	Implement region management	5.04 hrs
1.7.2	Associate states with regions	2.96 hrs
1.7.3	Associate EBDIs with states and regions	5.04 hrs
1.7.4	Implement regional queries	8 hrs

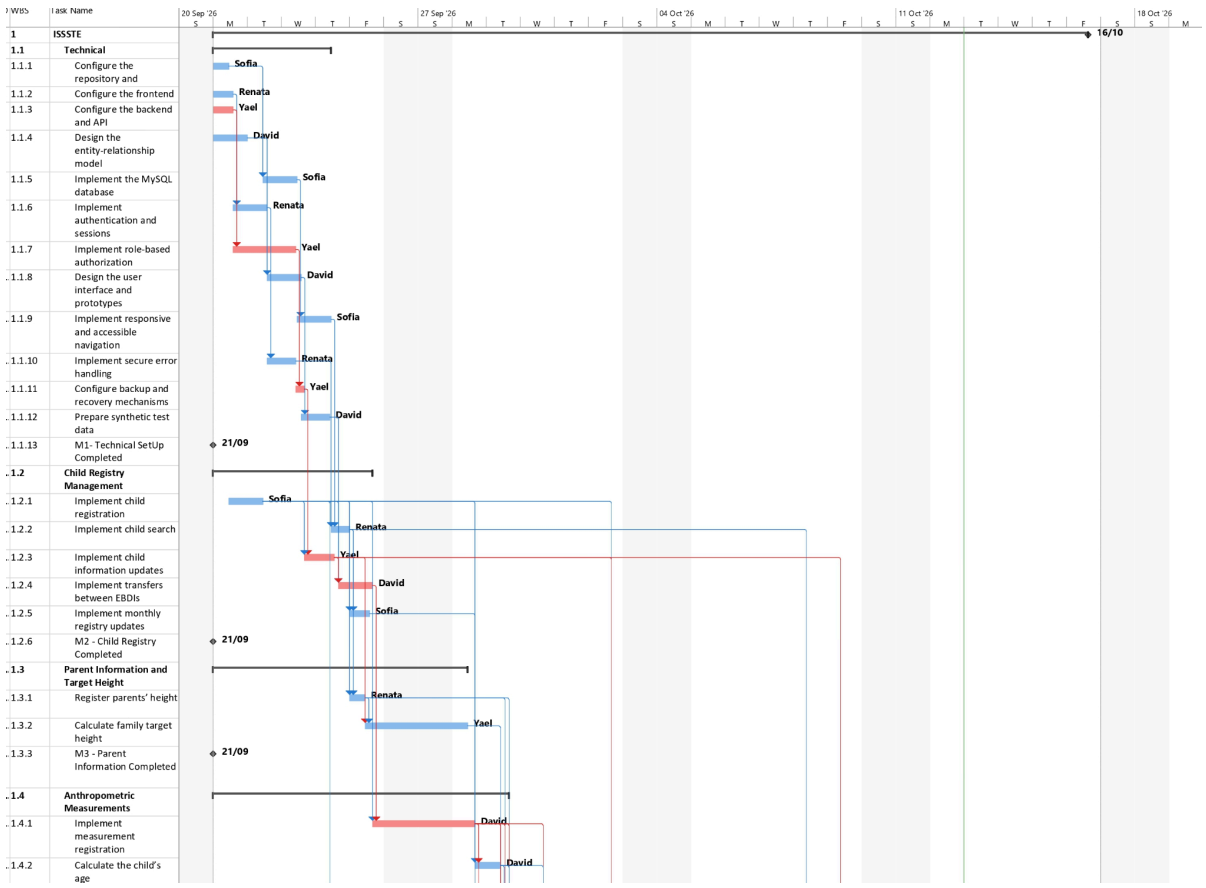
1.7.5	**Parent and Legal Guardian Access**	**23.12 hrs**
1.7.5.1	Associate guardians with children	5.04 hrs
1.7.5.2	Restrict guardian access	8 hrs
1.7.5.3	Implement result consultation for guardians	5.04 hrs
1.7.5.4	Protect registration number access	5.04 hrs
1.8	**Administration and Auditing**	**45.04 hrs**
1.8.1	Implement user management	8 hrs
1.8.2	Configure access scopes	8 hrs
1.8.3	Implement the audit log	8 hrs
1.8.4	Implement measurement correction	8 hrs
1.8.5	Implement report generation	13.04 hrs
1.9	**Testing and Delivery**	**94.24 hrs**
1.9.1	Design test cases	8 hrs
1.9.2	Execute unit tests	8 hrs
1.9.3	Execute integration tests	8 hrs
1.9.4	Validate clinical calculations	13.04 hrs
1.9.5	Execute security tests	13.04 hrs
1.9.6	Execute performance tests	5.04 hrs
1.9.7	Execute accessibility tests	5.04 hrs
1.9.8	Execute user acceptance testing	8 hrs
1.9.9	Develop the user manual	5.04 hrs

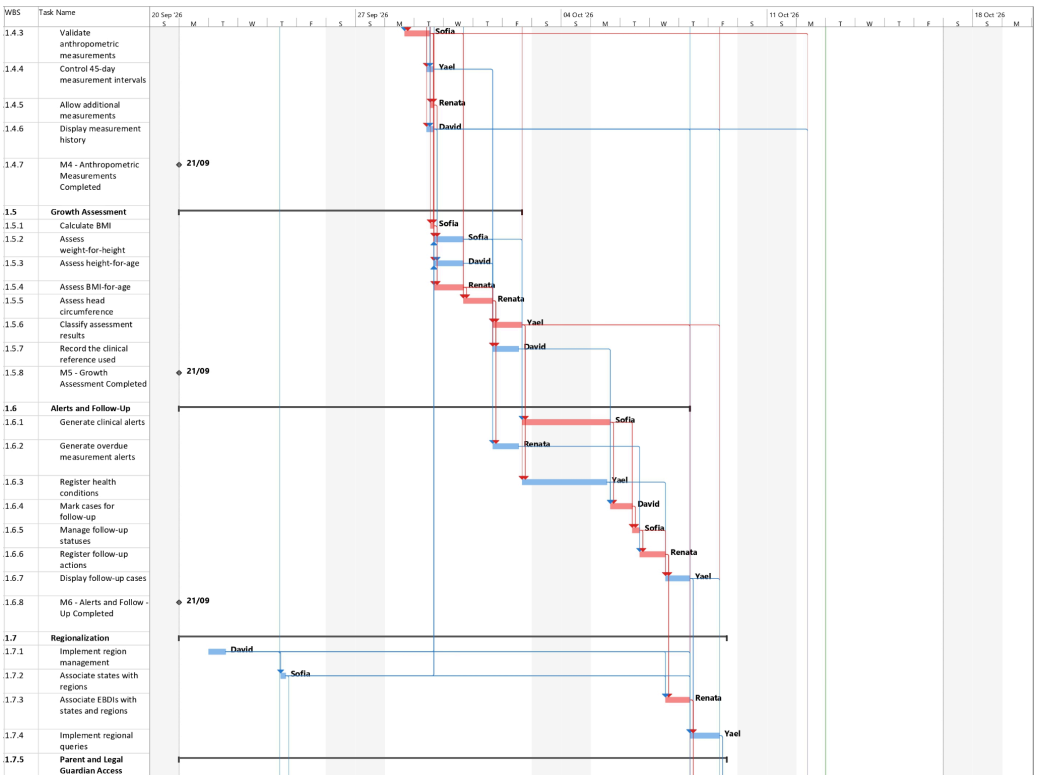
1.9.10	Document security test results	8 hrs
1.9.11	Develop the technical documentation	8 hrs
1.9.12	Conduct the demonstration and final delivery	5.04 hrs

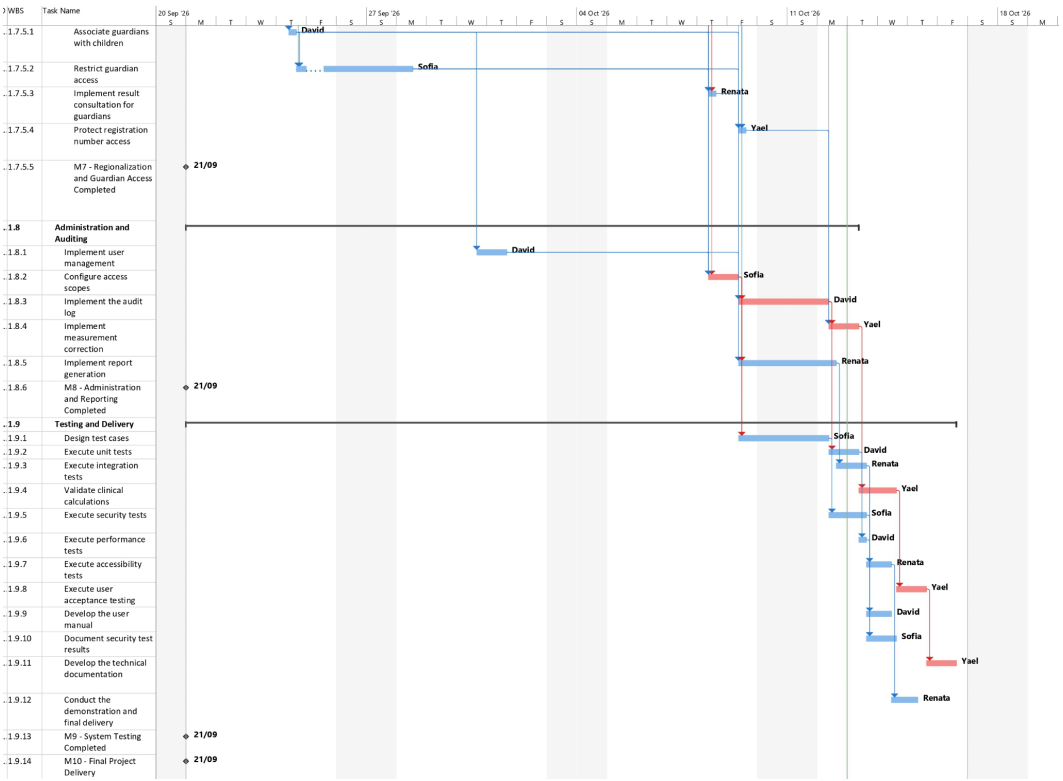
Schedule with milestones and critical path

The schedule was built in Microsoft Project from the WBS: each activity has an estimated duration and finish to start dependencies on the activities it needs as input.

Gantt Chart







Milestones

Milestones are zero duration checkpoints that mark the completion of each phase. Each one has a verifiable “done” criterion.

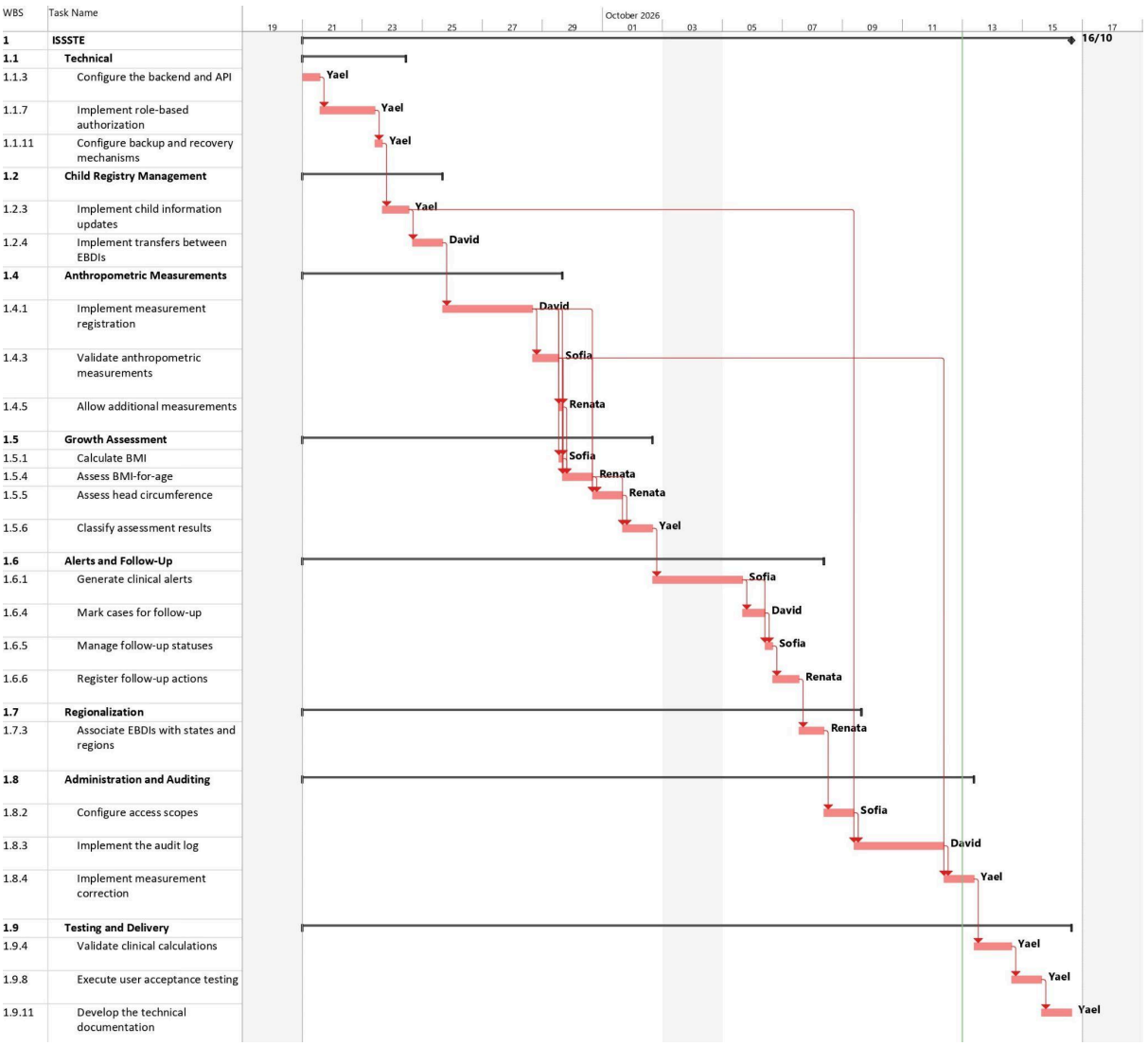
ID	Task Name	Closes phase	Done criterion
15	M1- Technical SetUp Completed	1.1	The frontend, backend, and database are running, and users can log in.
22	M2 - Child Registry Completed	1.2	Children can be registered, searched, updated and transferred between EBDIs.
26	M3 - Parent Information Completed	1.3	Parent’s height can be registered and the family target height is calculated.
34	M4 - Anthropometric Measurements Completed	1.4	Measurements can be registered, validated, and viewed in each child's history.
43	M5 - Growth Assessment Completed	1.5	The system calculates and classifies all growth indicators for a measurement.
52	M6 - Alerts and Follow - Up Completed	1.6	The system generates alerts and users can manage follow-up cases.
63	M7 - Regionalization and Guardian Access Completed	1.7	Queries work by region, and guardians can only see their own children.
70	M8 - Administration and Reporting Completed	1.8	Users can be managed, actions are recorded in the audit log, and

			reports are generated.
84	M9 - System Testing Completed	1.9	All tests are executed and the clinical calculations are validated.
85	M10 - Final Project Delivery	1.9	The system and its documentation are delivered and presented.

Critical Path

The critical path is the longest chain of dependent activities in the project. If any activity on this path is delayed, the whole project is delayed. It is calculated automatically by Microsoft Project from the durations and dependencies of the activities.

The critical path has 23 activities and lasts 19.78 days, which is the total duration of the project. It goes through almost every phase. Activities that are not on the critical path can be delayed a little bit without changing the final delivery date.



Human-resource assignment plan per activity

Each activity has one responsible team member. The assignment of every activity is shown in Microsoft Project, in the Resource Usage view below, where each team member appears with the list of activities assigned to them. The workload is balanced across the team, with between 96 and 110 hours per member. Yael has the most activities on the critical path, so his tasks will be continuously monitored to avoid delays.

Sofia	109.04 hrs	Work	8h	30.8h	20.16h	13.04h	14.8h	22.24h
Configure the repository and branching strategy	*2.96 hrs*	Work	2.96h					
Implement the MySQL database	*8 hrs*	Work		8h				
Implement responsive and accessible navigation	*8 hrs*	Work		8h				
Implement child registration	*8 hrs*	Work	5.04h	2.96h				
Implement monthly registry updates	*5.04 hrs*	Work		5.04h				
Validate anthropometric measurements	*5.04 hrs*	Work			5.04h			
Calculate BMI	*2.96 hrs*	Work			2.96h			
Assess weight-for-height	*8 hrs*	Work			8h			
Generate clinical alerts	*8 hrs*	Work				8h		
Manage follow-up statuses	*5.04 hrs*	Work				5.04h		
Associate states with regions	*2.96 hrs*	Work		2.96h				
Restrict guardian access	*8 hrs*	Work		3.84h	4.16h			
Configure access scopes	*8 hrs*	Work					8h	
Design test cases	*8 hrs*	Work					6.8h	1.2h
Execute security tests	*13.04 hrs*	Work						13.04h
Document security test results	*8 hrs*	Work						8h

Renata	96.32 hrs	Work	8h	18.08h	19.84h	5.04h	21.04h	24.32h
Configure the frontend	*5.04 hrs*	Work	5.04h					
Implement authentication	*8 hrs*	Work	2.96h	5.04h				
Implement secure error ha	*5.04 hrs*	Work		5.04h				
Implement child search	*5.04 hrs*	Work		5.04h				
Allow additional measuren	*2.96 hrs*	Work			2.96h			
Assess BMI-for-age	*8 hrs*	Work			8h			
Assess head circumference	*8 hrs*	Work			8h			
Generate overdue measur	*5.04 hrs*	Work			0.88h	4.16h		
Register follow-up actions	*5.04 hrs*	Work				0.88h	4.16h	
Associate EBDIs with state	*5.04 hrs*	Work					5.04h	
Implement result consultat	*5.04 hrs*	Work					5.04h	
Implement report generati	*13.04 hrs*	Work					6.8h	6.24h
Execute integration tests	*8 hrs*	Work						8h
Execute accessibility tests	*5.04 hrs*	Work						5.04h
Conduct the demonstratior	*5.04 hrs*	Work						5.04h
Register parents' height	*2.96 hrs*	Work		2.96h				

Yael	109.36 hrs	Work	8h	25.2h	8.88h	12.16h	18.08h	37.04h
Configure the backend ana	*5.04 hrs*	Work	5.04h					
Implement role-based autl	*13.04 hrs*	Work	2.96h	10.08h				
Configure backup and reco	*5.04 hrs*	Work		5.04h				
Implement child informatic	*5.04 hrs*	Work		5.04h				
Calculate family target hei	*8 hrs*	Work		5.04h	2.96h			
Control 45-day measureme	*5.04 hrs*	Work			5.04h			
Classify assessment results	*8 hrs*	Work			0.88h	7.12h		
Register health conditions	*5.04 hrs*	Work				5.04h		
Display follow-up cases	*5.04 hrs*	Work					5.04h	
Implement regional querie	*8 hrs*	Work					8h	
Protect registration numbe	*5.04 hrs*	Work					5.04h	
Implement measurement c	*8 hrs*	Work						8h
Validate clinical calculation	*13.04 hrs*	Work						13.04h
Execute user acceptance te	*8 hrs*	Work						8h
Develop the technical docu	*8 hrs*	Work						8h

David	105.2 hrs	Work	8h	32h	32h	7.12h	6.8h	19.28h
Design the entity-relations	*8 hrs*	Work	8h					
Design the user interface a	*8 hrs*	Work		8h				
Prepare synthetic test data	*5.04 hrs*	Work		5.04h				
Implement transfers betwe	*8 hrs*	Work		8h				
Implement measurement r	*8 hrs*	Work		0.88h	7.12h			
Assess height-for-age	*8 hrs*	Work			8h			
Record the clinical referenc	*5.04 hrs*	Work			0.88h	4.16h		
Mark cases for follow-up	*2.96 hrs*	Work				2.96h		
Implement region manage	*5.04 hrs*	Work		5.04h				
Associate guardians with c	*5.04 hrs*	Work		5.04h				
Implement user managem	*8 hrs*	Work			8h			
Implement the audit log	*8 hrs*	Work					6.8h	1.2h
Execute unit tests	*8 hrs*	Work						8h
Execute performance tests	*5.04 hrs*	Work						5.04h
Develop the user manual	*5.04 hrs*	Work						5.04h
Calculate the child's age	*2.96 hrs*	Work			2.96h			
Display measurement histo	*5.04 hrs*	Work			5.04h			

Risk Matrix

Each risk is scored on probability (P) and impact (I) on a scale from 1 to 5. The risk level is calculated as P X I and determines the attention the risk must receive.

Level (P X I)	Classification	Action
1 - 6	Low	Monitor
8 - 12	Medium	Plan
15 - 25	High	Act now

Risk Register

ID	Risk	P	I	Score	Level	Owner	Review
R1	Data loss	2	5	10	Medium	Sofía	At the end of the technical phase and before final delivery
R2	Unauthorized access to sensitive data	3	5	15	High	Renata	At the end of the technical phase and during security testing
R3	Errors in clinical calculations	3	5	15	High	Yael	At the end of the growth assessment phase and during testing
R4	ISSSTE delay in providing the clinical formulas and tables	5	4	20	High	Sofía	Weekly and before starting the growth assessment phase
R5	Incomplete or changing requirements	4	3	12	Medium	David	After each meeting with ISSSTE
R6	Team member unavailability	2	4	8	Medium	Renata	At each weekly team meeting
R7	Underestimated task durations	4	4	16	High	David	Weekly, comparing actual vs. planned progress

Probability x Impact Matrix

Probability/Impact	Low (1-2)	Medium (3)	High (4-5)
High (4-5)		R5	R4, R7
Medium (3)			R2, R3
Low (1-2)			R1, R6

All registered risks have medium or high impact because the system handles sensitive data and the delivery date is fixed.

Risk Details

R1. Data loss

- P = 2: the plan includes backups and version control from the technical phase, which reduces the probability of permanent data loss.
- I = 5: a child's growth history cannot be reconstructed, and without it, cases requiring follow up can't be detected.
- P X I = 10 (Medium)

Mitigation: schedule automatic weekly database backups, store the code and database scripts in the repository and never delete records directly, corrections preserve the original record.

Contingency: restore the data from the most recent backup and re-enter any data recorded after that backup using the EBDI's records.

R2. Unauthorized access to sensitive data

- P = 3: the system handles 4 profiles with different access type (EBDI, region, individual child), which requires careful permission control, which is already a known feature already included in the plan.
- I = 5: the data involved is personal and health information of minors, a leak would have legal consequences for ISSSTE and cause a loss of trust in the platform.
- P X I = 15 (high)

Mitigation: require a username/ID and password, verify user permissions on the server for every query, use encrypted connections (HTTPS), work only with synthetic data during development, and run security tests before delivery.

Contingency: block the affected accounts, review the audit log to identify what information was accessed, fix the vulnerability, and notify the ISSSTE.

R3. Errors in clinical calculations

- P = 3: multiple indicators are involved (BMI, weight-for-height, height-for-age, head circumference) with parameters that depend on age, sex, and region, and the formulas have not yet been validated.
- I = 5: an incorrect classification could result in a child with malnutrition or short stature not receiving follow-up, directly affecting their health.
- P x I = 15 (high)

Mitigation: create test cases calculated manually using CDC tables and compare them with the system's results, request that ISSSTE medical staff validate the calculations before delivery, and record the source and version of the table used in each assessment.

Contingency: disable alerts for the affected indicator and mark those assessments as "pending review", once the error is fixed, recalculate previous assessments. Since the system does not issue diagnoses, the final decision is always made by a person.

R4. ISSSTE delay in providing the clinical formulas and tables

- P = 5: the formulas and regional parameters have not been received, and ISSSTE has not set a delivery date.
- I = 4: the assessment calculations and alerts are on the critical path, and the regional parameters the ISSSTE requests are not available from any alternative public source, however, the CDC tables allow work to proceed with the alternative.
- P x I = 20 (High)

Mitigation: Formally request the formulas and regional parameters at the start of the project via email with a deadline, using professors as intermediary with the partner organization.

Contingency: Implement calculations using CDC tables as configurable parameters and mark results as not validated, and document a guide for updating the parameters once ISSSTE validates them.

R5. Incomplete or changing requirements

- P = 4: the only source of requirements is a brief document from ISSSTE, there are open questions in the SRS (regional coordinator permissions, who is responsible for follow-up), and communication with the partner is limited.

- I = 3: an incorrect assumption leads to rework in specific modules (permissions, interfaces, data model), not in the entire system.
- P x I = 12 (Medium)

Mitigation: document every assumption the team makes in the SRS, compile a list of open questions, and send it to ISSSTE (through the professors) to request written confirmation.

Contingency: If ISSSTE requests a change, the team evaluates the required effort, if it fits within the remaining time, it is implemented, otherwise, it is documented as future work.

R6. Team member unavailability

- P = 2: the team has shown consistent commitment, but workload from other courses, exams, or illness can affect any member's availability.
- I = 4: every team member has tasks on the critical path, the team can partially compensate by working on weekends.
- P x I = 8 (Medium)

Mitigation: maintain good communication through weekly progress meetings and give prior notice of exam weeks or deadlines in other courses, so the workload can be adjusted before delays occur.

Contingency: dedicate additional time, including weekends, and redistribute the absent member's pending tasks among the rest of the team, starting with those on the critical path.

R7. Underestimated task durations

- P = 4: the entire system is planned in 5 weeks, and it is the team's first time working with this stack.
- I = 4:since the delivery date is fixed, any delay directly translates into undelivered features, the impact is not rated a 5 because it is possible to reorganize to cover critical tasks.
- P x I= 16 (High)

Mitigation: break the work down into short tasks to detect delays early, and compare actual vs. planned progress every week.

Contingency: reassign team members to delayed tasks and, if that is not enough, deliver first the modules that others depend on and document the remaining work as future work.

Most Critical Risk

By score, the highest priority risk is R4. ISSSTE delay in providing the clinical formulas and tables with a score of 20, but it has a viable plan. In terms of the most damaging one, would be R2. Unauthorized access to sensitive data, because it is the only one whose damage cannot be undone since a child’s health data that has already been exposed cannot be recovered.